

Rayen Mrad

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🏠 Rayen Mrad | 🔄 fannen-prt1 | 🌐 My Portfolio | 📍 Monastir, Tunisia

EXPERIENCE

Welyne - Collab with AI SynapsePulse Inc. (USA)

Oct. 2025 - Jan. 2026

Web development and AI Intern

Tunis, Tunisia

- Conducted a research about the NFL systems , the sensors and data to use.
- Designed a platform to track NFL Players Metrics (HR, RR intervals , speed and accuracy) to predict their fatigue.
- Built a ML model to predict a one of the four levels of fatigue.
- Provided a quicker way to track players' fatigue to guarantee the best performance of a NFL team.

Get Wireless

June 2025 – July 2025

Image Processing Intern

Tunis, Tunisia

- Developed a Python script to enhance radiology image quality through improving quality techniques.
- Served a tool to help doctors track cancer by improving image quality.

Freelancing

2024

Online Freelancer

Remote

- Designed visual concepts for different clients based on their requirements.

Tunisie Telecom

2024

Summer Job

Monsatir ,Tunisia

- Assisted in technical and operational tasks within the telecom environment.
- Supported internal teams in handling daily service-related and system maintenance activities.

PROJECTS

PCB Design Project

Feb. 2026 – Apr. 2026

- Designed a printed circuit board (PCB) for an embedded electronic system.
- Developed schematics and PCB layout using standard electronic design tools.

Leak Detection System

Feb. 2026

- Developed an embedded system for leak detection using environmental sensors.
- Integrated sensor data with AI models to detect and classify leak events.
- Implemented real-time monitoring and alert mechanisms for system response.

Supply Chain Simulation System

Jan. 2026 – Feb. 2026

- Developed a supply chain simulation model using AnyLogic to replicate a production line system.
- Introduced perturbations and accident scenarios to analyze system robustness and performance under disruptions.
- Evaluated system behavior to study the impact of uncertainties on production flow and efficiency.

Machine Health Monitoring System

Oct. 2025 – Jan. 2026

- Collected and analyzed sensor datasheets to support system setup and integration.
- Developed a C++ program to simulate sensors and read real-time values.
- Integrated sensor data with AI models to predict machine health levels.

Automated Warehouse System Programming

Jan. 2025 – Mar. 2025

- Developed and simulated an automated production line using Factory I/O and PLCSIM.
- Implemented control logic in TIA Portal using Ladder Diagram programming for real-time operation.
- Worked with Siemens PLC simulation tools for industrial automation workflows.

Quality Control System

Oct. 2024 – Apr. 2025

- Developed a computer vision pipeline for industrial quality control and anomaly detection.
- Built a Streamlit dashboard for real-time anomaly monitoring and visualization.
- Achieved 93% accuracy for anomaly detection on the evaluated dataset.

Robotics Projects

Sept. 2022 – Present

- Built robots with basic functionalities (line following, all-terrain) from scratch.
- Designed electrical systems and 3D models to improve robot understanding and design.
- Evaluated robot performance and iteratively improved system behavior.

Extracurricular Activities

Competitive Programming Contests Staff

- Contributed to initiatives that promoted competitive programming culture and improved the quality of contests and learning activities in Tunisia.

Leader, Tunisian Scouts

- Developed leadership, team-building, and event coordination skills by leading scout activities and supporting group organization.

National Robotics Weekend, Organizing Committee Member

- Served as Administrative Affairs Manager, coordinating participant communication and administrative processes to ensure smooth event operations.

Robotics Workshops Instructor

- Delivered robotics workshops for the Tunisian Scouts during national events, including JOTA-JOTI and National Scout Day.

Instructor, Moataz Mdimagh Robotics Club

- Taught children foundational robotics concepts through simple, hands-on projects and accessible teaching methods.

General Secretary, ACM Club

- Managed meetings, activities, and internal communication to support members' programming growth and club engagement.

EDUCATION

National Institute of Applied Sciences and Technology (INSAT)

Sep. 2022 – Present

Industrial IT and Automation Engineering Degree

INSAT, Tunis

- **Coursework:** Embedded Systems, Automation, Electronics, Robotics.

SKILLS

Embedded Systems : Embedded C/C++, Electronics, Sensors & Actuators, Automation, Robotics, Real-Time Data Acquisition

Programming : C/C++, Python, Java, JavaScript/TypeScript, SQL

Tools : GitHub, SolidWorks, Fritzing, Proteus, Wokwi, VS Code

Microcontrollers : ESP32 , Arduino , STM32 ,RaspBerry Pi

Robotics Software Stack : ROS 2, Gazebo simulation environment